INCH-POUND
MIL-DTL-32086(AS)
9 April 2004

DETAIL SPECIFICATION

INTERCOM, PORTABLE

This specification is approved for use by the Naval Air Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the technical requirements for a body-worn, 9-volt D.C. battery-powered, portable intercom to be used with, and form part of, a Chemical, Biological, Radiological (CBR) protective assembly (see figure 1). The intercom described in this specification is designed for use with current and future military airborne communication systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ, 08733-5100 or emailed to Tom.Omara@navy.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-22442/7		Cable Assembly, Aircraft Audio Connector TJ-106

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD–810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment

(Copies of these documents are available on line at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Order Desk, 700 Robins Avenue, Building 4D, Philadelphia, PA 19111-5049.)

2.2.2 Other Government documents, drawings, and publication. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND

NAVAIR AD 1115	-	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
NAVAIR 13-1-6.7-1	-	Aviation-Crew Systems Aircrew Personnel Protective Equipment (Aircrew/Passenger Equipment)

(Unless otherwise indicated, copies of the above publications are available from the Commander, Naval Air Systems Command, Code AIR-7.2.5.3, Building 407, Naval Air Station, Patuxent River, MD 20670.)

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NAVSEA TE-00-AB-GTP-010 - Part Stress Derating

(Unless otherwise indicated, a copy of the above publication is available from the Commander, Naval Aviation Supply Office, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

THE INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS (IPC, INC.)

J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

(Applications for a copy should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 7380 North Lincoln Avenue, Lincolnwood, IL 60062-6135 or http://www.ipc.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Component parts. The intercom is comprised of a self-contained amplifier box with an internal 9-volt power source. The intercom shall provide the power and the electronics necessary to transfer audio signal from the intercom to either the headset or integral speaker. The intercom assembly shall be supplied with the following component parts:

1 Intercommunication Unit
1 Battery, 9V, Alkaline

3.3 Weight. The weight of the intercom (with battery) shall be not greater than 24 ounces.

3.4 Color. The intercom shall be colored black.

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3.5 Design. The design of the intercom shall meet the mechanical construction, electrical characteristics, and acoustical performance requirements of this specification. The intercom shall attach to components of the aircrew ensemble without modification of the garments. The intercom shall be for use both indoors and outdoors and while the user is donning and doffing his chemical, biological, radiological (CBR) protective ensemble. The intercommunication unit shall provide the user with listening and broadcasting capabilities to his immediate surroundings while leaving both hands free. Battery replacement shall be readily effected with CBR-gloved hands.

3.5.1 Dissimilar metals. Unless required, dissimilar metals as defined in MIL-STD-889 shall not be used in direct contact with each other.

3.5.2 Reverse polarity. The intercom shall not be damaged by the inadvertent reversal of the battery polarity.

3.5.3 Case. The case shall be smooth and contoured to prevent both the accumulations of liquid contaminants and abrasives or damage to other components of the aircrew ensemble. All corners shall be radiused and projections shall be avoided.

3.5.4 Battery housing. The battery housing shall be isolated and sealed from the environment and the other internal components of the intercom to prevent damage from battery leakage or venting.

3.5.5 Snap hook. The snap hook attaches to the case to allow connection to body-worn equipment.

3.5.6 Switch/Modes.

3.5.6.1 Switch. A switch shall be accessible and operable by the wearer utilizing a CBR-gloved hand. The switch shall enable the wearer to select the operational modes.

3.5.6.2 Operational modes. The intercom shall have the operational modes listed below (see figure 2).

Off - Communication between two operator positions with the intercom switched off.

Listen - Communication between the two operator positions together with the reception of signals from the intercom (operating in the microphone mode) at both headset positions.

Talk/Listen (Full Duplex) - Speech through the intercom (operating in the loudspeaker mode) from both operator positions.

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3.5.6.3 Mode markings. Switch position modes shall be marked with ink or a decal.

3.5.9 Battery. One 9-volt alkaline battery shall be used to supply electrical power for the intercom. The operational life of the battery shall be not less than six hours of continuous operation with a listen only-to-talk/listen ratio of 9:1.

3.5.10 Plugs and connectors/interface.

3.5.10.1 Plugs and connectors. One M22442/7-1 connector shall be located on the intercom via a 5.5 ±0.25 inch long cable (including the connector) and shall allow for the insertion of one mating communications cord.

3.5.10.2 Electrical interface. The intercom shall interface with the V-22 and CBR communications equipment as depicted on figure 3.

3.5.11 Dimensions. The dimensions of the intercom shall be as shown on figure 4. Configuration can vary as long as the unit stays within the maximum dimensions.

3.6 Construction.

3.6.1 Unacceptable Material. Natural rubber or natural rubber cement shall not be used. Rubber substitutes and plastic materials shall not be used as edges and fillers.

3.6.2 Intercom amplifier circuitry. All electronics shall be contained within the intercom assembly to minimize the risk of a fire potential. Voltages and currents in the intercom system shall not create sparks or any other adverse effect that presents a fire potential. This includes connecting and disconnecting any and all intercom cabling and connectors.

3.6.3 Cement. The cement used shall meet all test conditions.

3.6.4 Amplifier shield. To minimize the effects of electromagnetic interference or susceptibility, the amplifier shall be shielded.

3.6.5 Solderability. Soldering shall be performed in accordance with J-STD-001.

3.6.6 Aircrew safety. The construction shall be such that no potential voltages are present on any surface that could create a shock hazard to the user under any environmental condition.

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3.6.7 Cord, cordage, and cables.

3.6.7.1 Temperature range. The cable shall be flexible and resilient throughout the temperature range of -40 to +85 °C and shall show no evidence of cracking or any other damage as a result of exposure to the temperature range specified herein (see 4.5.3.1).

3.6.7.2 Flex life. The cable shall have a mean flex life of 25,000 flexes without showing evidence of the following (see 4.5.3.2):

a. Damage to the outside jacket (splits, cracks)
b. Damage to the insulation of the individual conductors
c. Electrical discontinuity

3.6.7.3 Cord anchorage. The cord and its anchorage between the connector plug and the unit body shall withstand an 8-pound pull applied for a period of 30 minutes without damage to the cord and its anchorage. Any slippage at the connector shall be considered a failure (see 4.5.3.3).

3.6.7.4 Conductor strain relief and twist relief. The conductors shall be anchored in the connector plug and unit body to provide strain relief for the conductors. Means shall be provided to prevent twisting or inward movement of the cord in the connector plug or unit body (see 4.5.3.4).

3.6.7.5 Insulation resistance. When tested in accordance with 4.5.3.5, the insulation resistance between the conductors and outside insulation shall be not less than 10 megohms.

3.6.7.6 Dielectric withstanding voltage. When tested as specified, the insulated conductors to the cable outside insulation shall withstand the application of 115 VAC nominal alternating potential of commercial line frequency without flashover or breakdown (see 4.5.3.6).

3.7 Performance.

3.7.1 Electrical characteristics.

3.7.1.1 Current consumption. Current draw shall be not more than 15 mA DC in the Listen mode and not more than 150 mA DC in the Talk/Listen mode when tested as specified (see 4.5.2.1.1).

3.7.1.2 Input impedance. The intercom circuit shall provide power and input impedance necessary for use with the M32015/5 CBR Mask Intercom assembly. The load at the intercom input shall be 150 ohms ± 5 percent (see 4.5.2.1.2).

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3.7.1.3 Frequency response. When tested as specified, the nominal audio frequency response of the intercommunication unit shall be as shown on figure 5 over the frequency range of 300 Hz to 3500 Hz (see 4.5.2.1.3).

3.7.1.6 Voltage operating range. The communication unit shall operate at 9 ± 1 Volts D.C. (see 4.5.2.1.4).

3.7.3 Acoustic.

3.7.3.1 Harmonic distortion. When tested as specified, the distortion shall be less than 5 percent (see 4.5.2.2.1).

3.7.3.2 Talk output level. The mean sound pressure level at 12 in. from the speaker unit shall be 80 dB ± 5 dB sound pressure level (SPL) at 1000 Hz with an applied input signal of 20 mV. The communication unit's Talk Output Level shall be within 3 dB of the initial Talk Output Level after subjection to environmental test conditions (see 4.5.2.2.2).

3.7.3.3 Listen level. The electrical output shall be between 330 mV to 570 mV with an applied sound pressure level (SPL) of 104 dB at 1000 Hz. The communication unit's Listen Level shall be within 3 dB of the initial listen level after subjection to environmental test conditions (see 4.5.2.2.3).

3.8 Environmental.

3.8.1 High temperature. When tested as specified in 4.5.4.1, the intercom shall operate without failure.

3.8.2 Low temperature. When tested as specified in 4.5.4.2, the intercom shall operate without failure.

3.8.3 Humidity. Upon completion of the humidity test, the intercom shall meet the requirements of 3.7.3.2 and 3.7.3.3 (see 4.5.4.3).

3.8.4 Vibration. Upon completion of the vibration test, the intercom shall meet the requirements of 3.7.3.2 and 3.7.3.3 (see 4.5.4.4).

3.8.5 Shock. Upon completion of the shock test, the intercom shall meet the requirements of 3.7.3.2 and 3.7.3.3. There shall be no evidence of breaking, cracking, or other structural damage (see 4.5.4.5).

3.8.6 Electromagnetic environmental effects. When tested as specified, the intercom shall operate without failure and show no signs of radiated susceptibility (see 4.5.4.6).

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3.8.7 Salt fog. When tested as specified, the intercom shall operate without failure and show no structural degradation (see 4.5.4.7).

3.8.8 Rain. When tested as specified, the intercom shall operate without failure (see 4.5.4.8).

3.8.9 Fungus. When tested as specified, the intercom shall not support fungal growth and shall operate without failure (see 4.5.4.9).

3.8.10 Low pressure (altitude). When tested as specified in 4.5.4.10, the intercom shall operate without failure.

3.8.11 Dust. When tested as specified, the intercom shall operate without failure (see 4.5.4.11).

3.9 Identification of product.

3.9.2 Marking. A label shall be affixed to the unit and contain the following information:

M32086 Intercom, Portable
Manufacturer's Name and/or CAGE Code
Contract Number
Date Code
Serial Number.

3.10 Environmental stress screening (ESS). When specified in the contract, the contractor shall subject each intercom to, and successfully pass ESS at the completion of quality conformance inspection and prior to delivery.

3.11 Reliability. The intercommunication unit shall meet a reliability minimum of $?_0$ = 5000 hours with a listen only-to-talk/listen ratio of 9:1.

3.12 Workmanship. The intercom shall be manufactured in a manner that results in a product of uniformly excellent quality, free from defects that could effect safety, performance, reliability, or durability. The intercom shall be free from contamination and imperfections such as dirt, grease, oil, foreign matter, malfunctioning of hardware and damage, such as cracks, discoloration, stress flaws, and sharp edges.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in Section 4 of this specification.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities with the accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.2.2 Test conditions. Unless otherwise specified (see 6.2), the intercom shall be tested under the following conditions.

a. Temperature: Room ambient. (+l5 to +35 °C)
b. Pressure: Normal atmosphere
c. Humidity: Prevailing ambient up to 90 percent relative humidity

4.2.3 Plotting of frequency response characteristic. The frequency response characteristic of the intercom, if plotted, shall be plotted on a graphic level recorder using semilogarithmic coordinate paper with the response in dB on the linear ordinate scale and the frequency in Hz on the logarithmic abscissa scale. The length of a 100-to-1 frequency interval on the abscissa scale shall equal the length of 50 dB on the ordinate scale.

4. 3 First article inspection. The contractor shall perform first article inspection after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units, which have been produced with equipment and procedures normally used in production.

4. 3.1 Sample size. Five intercoms shall be subjected to first article inspection.

4.3.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. All sample units shall be subjected to the inspections of group I. The samples shall then be divided into two groups and subjected to the inspections for their particular group (for groups II and III).

4.3.3 Failures. More than one failure in group I and any failure in group II or III shall be causes for refusal to grant first article approval.

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TABLE I. First article inspection.

Inspection (units)	Requirement Paragraph	Test method Paragraph
Group I (5 samples)		
Visual and mechanical and inspection	3.2, 3.3, 3.4, 3.5, 3.6, 3.9, and 3.12	4.5.1
Current consumption *	3.7.1.1	4.5.2.1.1
Impedance *	3.7.1.2	4.5.2.1.2
Frequency response *	3.7.1.3	4.5.2.1.3
Voltage operating range *	3.7.1.6	4.5.2.1.4
Harmonic distortion	3.7.3.1	4.5.2.2.1
Talk output level *	3.7.3.2	4.5.2.2.2
Listen level *	3.7.3.3	4.5.2.2.3
Group II (1 sample)		
Temperature range	3.6.7.1	4.5.3.1
Flex life	3.6.7.2	4.5.3.2
Cord anchorage	3.6.7.3	4.5.3.3
Conductor strain relief and twist relief	3.6.7.4	4.5.3.4
Insulation resistance	3.6.7.5	4.5.3.5
Dielectric withstanding voltage	3.6.7.6	4.5.3.6

* Indicates minimum tests for First Article Testing.

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TABLE I. First article inspection. - Continued.

Inspection	Requirement Paragraph	Test Method Paragraph
Group III (4 samples)		
Environmental Subgroup A (2 samples)		
High temperature	3.8.1	4.5.4.1
Low temperature	3.8.2	4.5.4.2
Humidity *	3.8.3	4.5.4.3
Vibration	3.8.4	4.5.4.4
Low pressure (altitude)	3.8.10	4.5.4.10
Subgroup B (1 samples)		
Electromagnetic environmental effects (E3)	3.8.6	4.5.4.6
Salt fog *	3.8.7	4.5.4.7
Subgroup C (1 sample)		
Shock *	3.8.5	4.5.4.5
Rain	3.8.8	4.5.4.8
Fungus	3.8.9	4.5.4.9
Dust	3.8.11	4.5.4.11

* Indicates minimum tests for First Article Testing.

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4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.4.1.4.4, delivery of products, which have passed groups A and B inspections shall not be delayed pending the results of group C inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all intercoms of the same type, produced under the same conditions, and offered for inspection at one time.

4.4.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, in the order shown.

4.4.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table III. If one or more defects are found the lot shall be re-screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table III. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE II. Group A inspection.

Group A Inspection	Requirement Paragraph	Test Method Paragraph
Visual and mechanical inspection	3.2, 3.3, 3.4, 3.5, 3.6, 3.9, and 3.12	4.5.1
Impedance	3.7.1.2	4.5.2.1.2
Talk output level	3.7.3.2	4.5.2.2.2
Listen level	3.7.3.3	4.5.2.2.3

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TABLE III. Group A. zero defect sampling plan.

LOT SIZE	SAMPLE SIZE		
	ELECTRICAL CHARACTERISTICS	VISUAL and MECHANICAL EXAMINATION	
		MAJOR	MINOR
2 to 8	100 percent	100 percent	3 percent
9 to 15	100 percent	13	3
16 to 25	100 percent	13	3
26 to 50	100 percent	13	5
51 to 90	100 percent	13	6
91 to 150	125	13	7
15l to 280	125	13	10
281 to 500	125	29	11
501 to 1,200	125	34	15
1,201 to 3,200	125	42	18
3,201 to 10,000	294	50	22
10,001 to 35,000	294	60	29

Note: If sample size equals or exceeds lot size, 100 percent inspection is required.

4.4.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IV and shall be made on sample units that have been subjected to, and passed group A inspection.

4.4.1.3.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table V. If one or more defects are found the lot shall be re-screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table V. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.4.1.3.2 Disposition of samples. Sample units that have passed the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified electrical tolerances.

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TABLE IV. Group B inspection.

Inspection	Requirement Paragraph	Test Method Paragraph
Current consumption	3.7.1.1	4.5.2.1.1
Frequency response	3.7.1.3	4.5.2.1.3
Voltage operating range	3.7.1.6	4.5.2.1.4
Harmonic distortion	3.7.3.1	4.5.2.2.1
Insulation resistance	3.6.7.5	4.5.3.5
Dielectric withstanding voltage	3.6.7.6	4.5.3.6
Flexing life	3.6.7.2	4.5.3.2
Cord anchorage	3.6.7.3	4.5.3.3
Conductor strain relief and twist relief	3.6.7.4	4.5.3.4

TABLE V. Group B. zero defect sampling plan.

LOT SIZE	SAMPLE SIZE
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
15l to 280	9
281 to 500	11
501 to 1,200	13
1,201 to 3,200	15
3,201 to 10,000	15
10,001 to 35,000	15

NOTE: If sample size equals or exceeds lot size, 100 percent inspection is required.

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4.4.1.4 Group C inspection. Group C inspection shall consist of the tests specified in table VI in the order shown. Group C inspection shall be made on sample units that have passed groups A and B inspections.

TABLE VI. Group C inspection.

Inspection	Requirement Paragraph	Test Method Paragraph
Subgroup 1 (5 sample units)		
Sub-Subgroup A (2 samples)		
High temperature	3.8.1	4.5.4.1
Low temperature	3.8.2	4.5.4.2
Humidity	3.8.3	4.5.4.3
Sub-Subgroup B (1 sample)		
Vibration	3.8.4	4.5.4.4
Rain	3.8.8	4.5.4.8
Low pressure (altitude)	3.8.10	4.5.4.10
Sub-Subgroup C (1 sample)		
Salt fog	3.8.7	4.5.4.7
Shock-drop	3.8.5	4.5.4.5
Subgroup 2 (1 sample units)		
Dust	3.8.11	4.5.4.11

4.4.1.4.1 Sampling plan.

4.4.1.4.1.1 Sample size. Five samples of the complete intercom assembly shall be selected at random; four samples shall be assigned for subgroup l and one sample for subgroup 2 of group C in table VI. The samples shall be selected at the start of the contract from the first quality conformance inspection lot. These samples shall constitute the group C requirement for the first 1,000 units produced. Thereafter, five samples of the complete assemblies shall be selected at random for subgroup l of group C in table VI. These samples shall be selected once each month, or every 1,000 units, whichever occurs first.

4.4.1.4.2 Failures. If one or more sample units fail to pass group inspection, the lot shall be considered to have failed.

4.4.1.4.3 Disposition of sample. Sample units that have been subjected to group C inspection shall not be delivered on the contract or purchase order.

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4.4.1.4.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the contracting activity and the cognizant inspection activity of such failure. The manufacturer shall then take corrective action on the materials or processes (or both) as warranted. The manufacturer shall also take corrective action on all units of product that can be corrected, that were manufactured essentially with the same materials and the same processes, and that are subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the cognizant inspection activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (on all inspections or on the inspection that the original sample failed, at the option of the procuring activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity.

4.4.1.4.5 Defects in items already accepted. The investigation of a test failure could indicate that defects might exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and the method of correcting them.

4.5 Methods of inspection.

4.5.l Visual and mechanical inspection. The intercom shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4, 3.5, 3.6, 3.7, 3.9, and 3.11). The defects shall be classified as specified in table VII. Particular attention shall be paid to the following where applicable.

a. Soldering operations
b. Uniformity and strength of cementing
c. Security and cleanliness of intercom terminal connections to cable terminals
d. Construction of intercom cord assembly
e. Vent in intercom case for pressure equalization
f. Cleanliness of vent holes throughout the intercom structure
g. Workmanship regarding the fit and sealing of the intercom case
h. Set screws holding the intercom plug to the intercom (Slotted, headless screws shall be used)
i. Weight.

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TABLE VII. Classification of defects for visual and mechanical inspection.

Defect Type	Classification	
	Major	Minor
Dimensions Materials and finish	Dimensions not as specified. Materials not as specified. Wrong or incomplete finish. Large amounts of flaking, peeling, or chipping of finish.	Scratches, cuts, abrasions, and similar defects, causing exposure of base metal or relatively small amounts of flaking, peeling, or chipping.
Parts	Missing parts. Inoperative, improperly assembled, or defective parts that could cause the unit to fail in service. Wrong parts.	Defective parts that would reduce efficiency of use, but not cause failure in service. Cracks or chipped surfaces not having an effect on the functioning, assembly, maintenance, or life of the unit.
Marking	Marking missing, illegible, or incorrect.	Marking dirty or smudged but legible
Foreign objects	Any metallic foreign object not firmly attached that could cause a short circuit or acoustical malfunctioning of the unit. Any nonmetallic foreign object such as insulation, dirt, or phenolic chips that could cause acoustical malfunctioning of the unit.	Any metallic or nonmetallic foreign object that affects appearance but could not cause acoustical malfunctioning of the unit.
Soldering	Improper wrap- Less than l/2 turn. Unsoldered joint- Solder not applied where intended. Insufficient solder-Minimum dimensions of solder bridge less than twice the diameter of the wire or less than 3/32 inch, whichever is greater. Entire area of contact between wire and terminal not jointed by solder bridge.	Improper wrap-more than l turn. Excess solder-Buildup solder on joint greater than necessary for good soldering, usually resulting in obliteration of wire contour.

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TABLE VII. Classification of defects for visual and mechanical inspection. Continued.

Defect type	Classification	
	Major	Minor
Soldering	Cold solder joint-Chalky appearance, lacks metallic luster, presents rough "pileup" appearance, movement of wire or solder upon pick application. Rosin joint-Presence of excess rosin, relative movement of wire or solder upon pick application. Insulation in terminal hole - Solder over insulation, no appearances of visible wire contour.	Cold solder joint - Chalky appearance, lacks metallic luster, presents rough "pile-up" appearance; no relative action between wire and solder upon pick application
Wiring	Wiring not in accordance with schematic diagram. Broken strands more than 20 percent. In a 7-strand conductor, more than 2 broken strands. Insulation burned, abraded, pinched, or deteriorated between two or more conductors resulting in a potential short circuit. Taut wire-Wire exhibits no slack and subsequent breakage may occur due to stress on terminal parts. Insulation frayed to the extent that a potential short circuit exists.	Broken strands-20 percent or less. In a 7 strand conductor, 2 broken strands or less Insulation burned, abraded, pinched, or deteriorated, with exposure of bare wire, but short circuit not possible Taut wire-Slight stress on conductor but no possibility of subsequent breakage.

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4.5.2 Performance.

4.5.2.1 Electrical.

4.5.2.1.1 Current consumption. With the proper intercom/receiver connected, the intercommunication unit shall be turned to the listen position and the current measured with a DC current meter. The intercommunication unit shall meet specified requirements. The intercommunication unit shall then be turned to the talk/listen position and the current measured with a DC current meter. The intercommunication unit shall meet specified requirements (see 3.7.1.1).

4.5.2.1.2 Input impedance. A 20 mV unloaded signal from a signal generator with 150-ohm output impedance shall be applied to the intercom input. The signal level shall be reduced to 10 mV ± 20 percent (8-12 mV)(see 3.7.1.2).

4.5.2.1.3 Frequency response. With an input voltage of 10 mV into the terminals of either jack socket, the audio frequency response of the amplifier shall be tested over the frequency range of 300 Hz to 3500 Hz. The intercom shall meet specified requirements (see 3.7.1.3).

4.5.2.1.4 Voltage operating range. The intercommunication unit shall meet the requirements of 3.7.3.2 and 3.7.3.3 when the input voltage source is adjusted between 8 VDC and 10 VDC (see 3.7.1.6).

4.5.2.2 Acoustic.

4.5.2.2.1 Harmonic distortion. With an input signal voltage of 10 mV, the harmonic distortion shall be less than 5 percent @ 1000 Hz (see 3.7.3.1).

4.5.2.2.2 Talk output level. With an input voltage of 20 mV, measure the acoustical output at 1000 Hz. The intercom shall meet the requirements specified in 3.7.3.2 at a distance of 12 inches coaxial from the loudspeaker.

4.5.2.2.3 Listen level. A sound pressure level of 104 dB shall be applied at the face of the transducer. The mean electrical output into 600 ohms over the frequency band 300 Hz to 3500 Hz shall meet the requirements specified in 3.7.3.3.

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4.5.3 Cord, cordage, and cable.

4.5.3.1 Temperature range. A sample of unterminated cable, approximately 1.5 feet long, shall be subjected to the following test. The unterminated cable shall be placed in a test chamber at temperature of +85 ±1 °C for a period of four hours. The cable shall be wrapped around a mandrel while exposed to this temperature without damage to the cable jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cable diameter. The cable shall be attached to the horizontal mandrel and suspended vertically, with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping around the mandrel without handling, by rotating the mandrel. The cable shall be inspected for damage or deformation and then placed in a test chamber at a temperature of -40 ±1 °C for 20 hours. The cable shall then be wrapped about the mandrel in the same manner as it was for the high temperature test. The cable shall show no cracks or other damage to the jacket or conductor insulation (see 3.6.7.1).

4.5.3.2 Flex life. Five samples of unterminated cable, each approximately 1.5 feet long, shall be subjected to the following test. Other flex test methods that will subject the cable to flex cycles equivalent to the flex cycles specified herein may be used upon approval by the contracting activity. The sample lengths shall be clamped and suspended through holes in a rectangular metal bar measuring 0.5 x 0.625 inches in the manner specified below (see 3.6.7.2).

a. The cable shall be threaded through the 0.625-inch dimension of the bar. The diameter of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cable undergoing test.

b. The cable shall be clamped immediately above the metal bar such that the cable clamp shall rest on top of the metal bar and shall not allow the cable to rotate in the bar.

c. The through holes in the metal bar shall incorporate a radius of 0.125 inch at the end through which the cable emerges (bending point).

d. A weight of 0.5 pound shall be hung from the cable.

e. The two conductors plus the shield of each sample length of cable shall be connected in series to a 6V/150 ma lamp, and a power supply to indicate electrical continuity.

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f. The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetic average of the number of flexes for each of the five sample lengths of cable before electrical discontinuity occurs shall be taken as the flex life of the cable.

4.5.3.3 Cord anchorage. With the connector firmly fixed, a pull of 8 pounds shall be applied to the cable in the direction of the axis of the extended intercom connector for a minimum of 30 minutes. There shall be no slippage of the connector or the intercom case.

4.5.3.4 Conductor strain relief and twist. With the intercom body anchored rigidly in a vertical position and the cable/connector extended, a 5-pound weight shall be suspended from the connector of the cord assembly for a period of at least 30 minutes. During the last 5 minutes of the test, the connector with the 5-pound weight suspended shall be twisted three turns clockwise and three turns counterclockwise from the normal resting position. The twisting cycle shall be repeated three times. The impedance of the intercom with cord assembly in place shall be measured at 1,000 Hz and a mechanical inspection shall be made before and immediately upon completion of the test. Upon completion of this test, a dielectric withstanding voltage test shall be made on the cord assembly in accordance with the procedure specified in 4.5.3.6 after the impedance measurement and mechanical inspection have been completed (see 3.7.1.2 and 3.5).

4.5.3.5 Insulation resistance. The insulated cable shall be tested in accordance with MIL-STD-202, Method 302. There shall be no evidence of arcing or insulation breakdown during the test. The insulation resistance shall be as specified (see 3.6.7.5).

4.5.3.6 Dielectric withstanding voltage. The insulated cable shall be tested in accordance with MIL-STD-202, Method 301 (see 3.6.7.6). The following apply:

a. Test voltage: 115 volts rms.

b. Nature of potential: 60 Hz AC.

c. Points of application – between one terminal and the external surface of the side cable insulation.

4.5.4 Environmental. Environmental tests shall be conducted in the order listed below. Prior to and at the completion of each environmental test, the intercom shall be tested as specified in 4.5.2.2.2 and 4.5.2.2.3 and shall meet the requirements specified in 3.7.3.2 and 3.7.3.3.

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4.5.4.1 High temperature. The intercom shall be tested as specified in MIL-STD-810, Method 501.4, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be raised to 71 °C at a rate of 3 °C per minute. The temperature shall be maintained for a period of two hours. The temperature shall then be brought to 55 °C at a rate of 3 °C per minute and maintained for a period of two hours. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 during and after the high temperature exposure.

4.5.4.2 Low temperature. The intercom shall be tested as specified in MIL-STD-810, Method 502.4, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be lowered to -40 °C at a rate of 3 °C per minute. The temperature shall be maintained for a period of two hours. The temperature shall then be brought to -25 °C at a rate of 3 °C per minute and maintained for a period of two hours. The intercom shall meet the requirements specified 3.7.3.2 and 3.7.3.3 during and after the low temperature exposure.

4.5.4.3 Humidity. See 3.8.3. The intercommunication unit shall be tested in accordance with MIL-STD-810, Method 507.4, except as follows:

Step 1. The chamber temperature shall be increased to +60 °C and the RH increased to 95 percent in 2 hours and maintained for 6 hours.

Step 2. Reduce internal chamber temperature in 4 hours to 28 ±10 °C and maintain 85 percent relative humidity for 12 hours.

Step 3. Repeat steps 1 and 2 for a total of 10 cycles (not less than 240 hours).

Step 4. The intercom shall be operated within one hour following completion of the 10th cycle, and tested to verify compliance with (3.7.3.2) and (3.7.3.3).

4.5.4.4 Vibration. The intercom shall be tested in accordance with Method 514.5, Procedure I of MIL-STD-810. The intercom unit shall be attached to the vibration table. The unit shall be subjected to sinusoidal vibration over a frequency range of 5 to 20 Hz at constant displacement double-amplitude of 0.10 inch, and over a frequency range of 20 to 500 Hz at a constant acceleration of 2g. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range. One vibration cycle (5-500-5) shall consume 15 minutes. Total time of the test shall be two hours. Repeat the above vibration for each axis. After test, the unit shall comply with 3.7.3.2 and 3.7.3.3 (see 3.8.4).

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4.5.4.5 Shock. The intercom shall be tested in accordance with Method 516.5, Procedure II of MIL-STD-810, with the following exceptions. For each of the three planar directions, the intercommunication unit shall be subjected to 4 shocks of 25g (245 m/s^2). Each effective shock pulse shall have a duration of 6 ms. The intercom shall meet the requirements of 3.7.3.2 and 3.7.3.3 (see 3.8.5).

4.5.4.6 Electromagnetic environment effects (E3). When subjected to the electric field strengths listed below, the system shall not malfunction, exhibit undesirable response beyond its performance tolerances, nor shall it produce acoustic noise greater than 75 dB(A) SPL. These EMI tests shall be performed in accordance with MIL-STD-461 (see 3.8.6). For acquisition of data in support of these tests (see 6.2).

TABLE VIII. Electric field strengths.

	FREQUENCY RANGE	FIELD INTENSITY	MODULATION
1.	14 kHz to 2 MHz	1 V/m	AM 50%, 1 kHz tone
2.	2 MHz to 30 MHz	10 V/m	AM 50%, 1 kHz tone
3.	30 MHz to 2 GHz	5 V/m	AM 50%, 1 kHz tone

4.5.4.7 Salt fog. The intercom shall be tested as specified in MIL-STD-810, Method 509.4. The test shall consist of four alternating periods of 24 hours exposure and 24 hours drying time. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 (see 3.8.7).

4.5.4.8 Rain. The intercom shall be tested as specified in MIL-STD-810, Method 506.4, Procedure I, Blowing Rain, for a period of one hour at a rate of four inches per hour. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 (see 3.8.8).

4.5.4.9 Fungus. The intercom shall be tested as specified in MIL-STD-810, Method 508.5 for a period of 28 days. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 (see 3.8.9).

4.5.4.10 Low pressure (altitude). The intercom shall be tested as specified in MIL-STD-810, Method 500.4, Procedure II, Operation. The intercom shall be placed in a climatic chamber. The chamber temperature shall be reduced to –25 °C at a rate not exceeding 3 °C per minute and the condition shall be maintained for a period of 30 minutes. The pressure in the chamber shall then be reduced to 15,000 feet of altitude over a period of 15 minutes and maintained at this pressure for 30 minutes. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 during exposure (see 3.8.10).

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4.5.4.11 Dust. The intercom shall be tested as specified in MIL-STD-810, Method 510.4, Procedure I, Blowing Dust for a period of six hours at 23 °C and 300 feet per minute air velocity and six hours at 55 °C and 1750 feet per minute air velocity. Relativity humidity shall be not greater than 30 percent. The intercom shall meet the requirements specified in 3.7.3.2 and 3.7.3.3 (see 3.8.11).

4.5.5 Reliability. When required by the contract, the contractor shall perform detailed design reliability analysis. It is recommended that the part stress analysis method be in accordance with MIL-HDBK-217 for airborne inhabited transport. Derating of parts shall be performed in accordance with TE-00-AB-GTP-010. The results of the analysis shall be documented in accordance with the provisions of the contract.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.l Intended use. The intercom covered by this specification is intended to provide the wearer of military unique protective assemblies in CBR contamination environments with the ability to converse:

a. with another similarly dressed person.

b. with the wearer of a respirator with no intercom/telephone equipment.

c. with an unprotected person.

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6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of the specification

b. If first article inspection is required (see 3.1)

c. Environmental stress screening (ESS) (see 3.10)

d. EMI test plan (see 4.5.5.8.1)

e. Reliability analysis if required (see 4.5.6)

d. Packaging required (see 5.1)

f. First article test program (see 6.3).

g. First article test plan (see 6.3)

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, or a standard production item from the contractor's current inventory. The number of items to be tested is specified in 4.4.l. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or identifying numbers (PIN). Part number for intercoms will be M32086-1.

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6.5 Subject term (key word) listing.

Acoustic
Aircrewman
Biological
CBR protective
Chemical
Distortion
Frequency response
Harmonic distortion
High gain
Intercom
Radiological

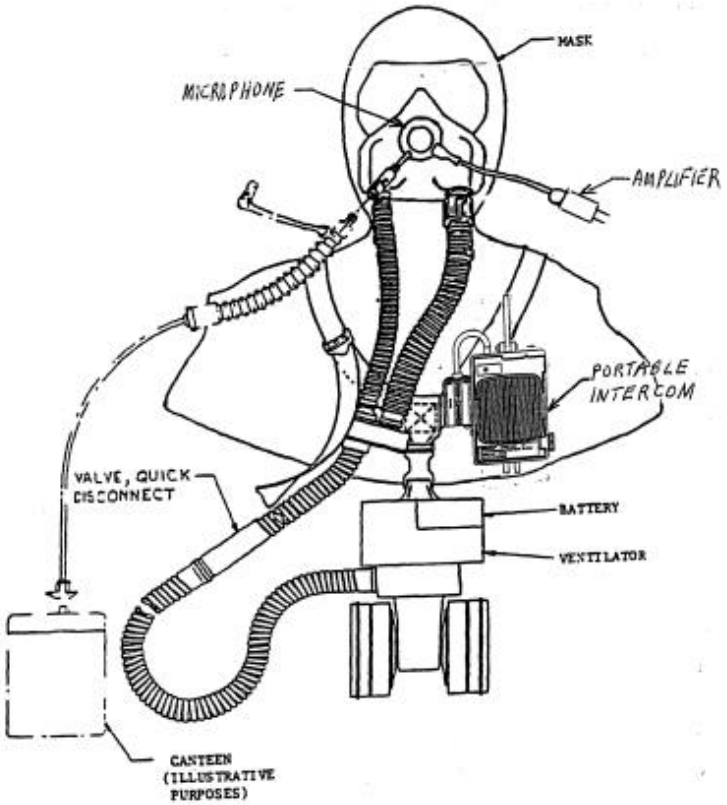


FIGURE 1. **Protective assembly, CBR, helicopter aircrewmember.**

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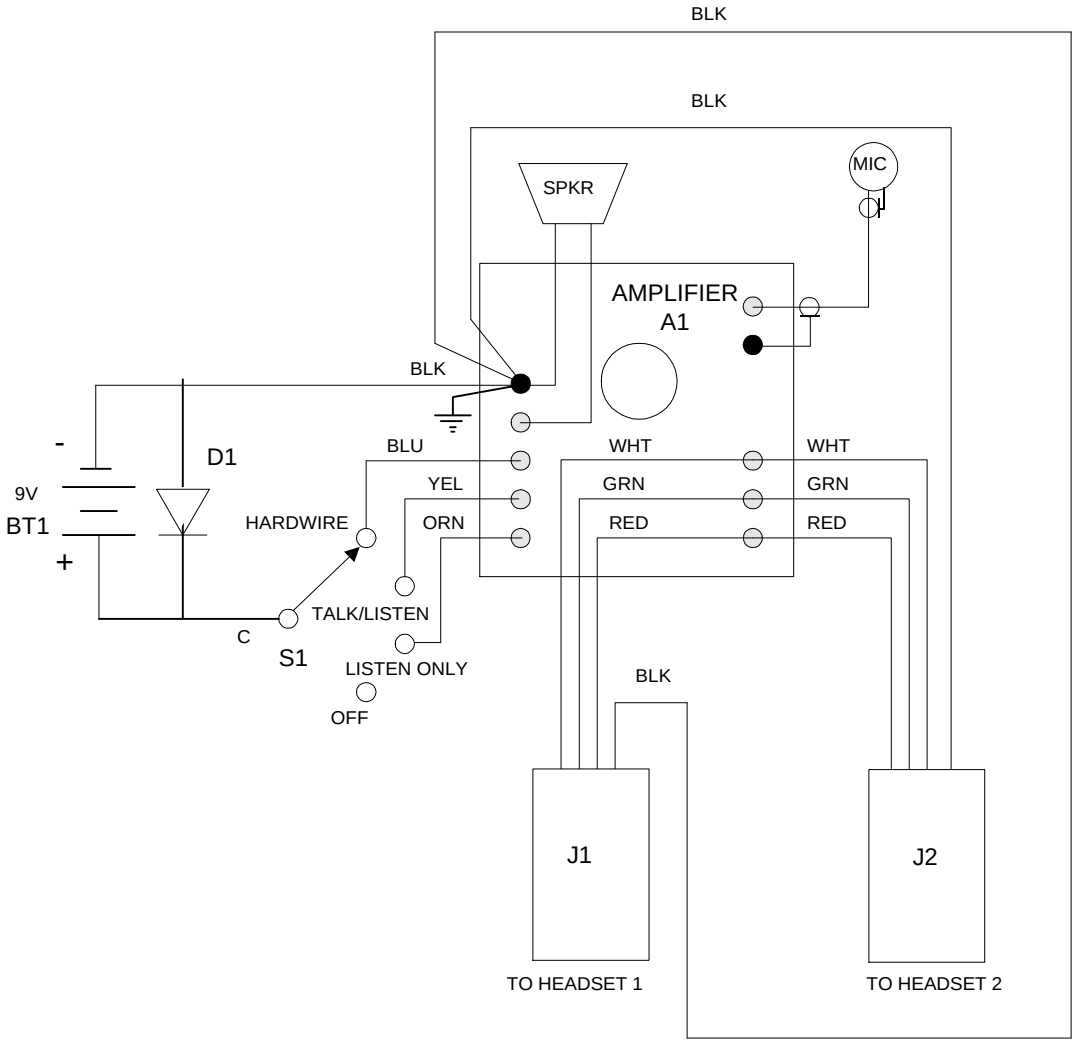


FIGURE 2. **Intercom block diagram.**

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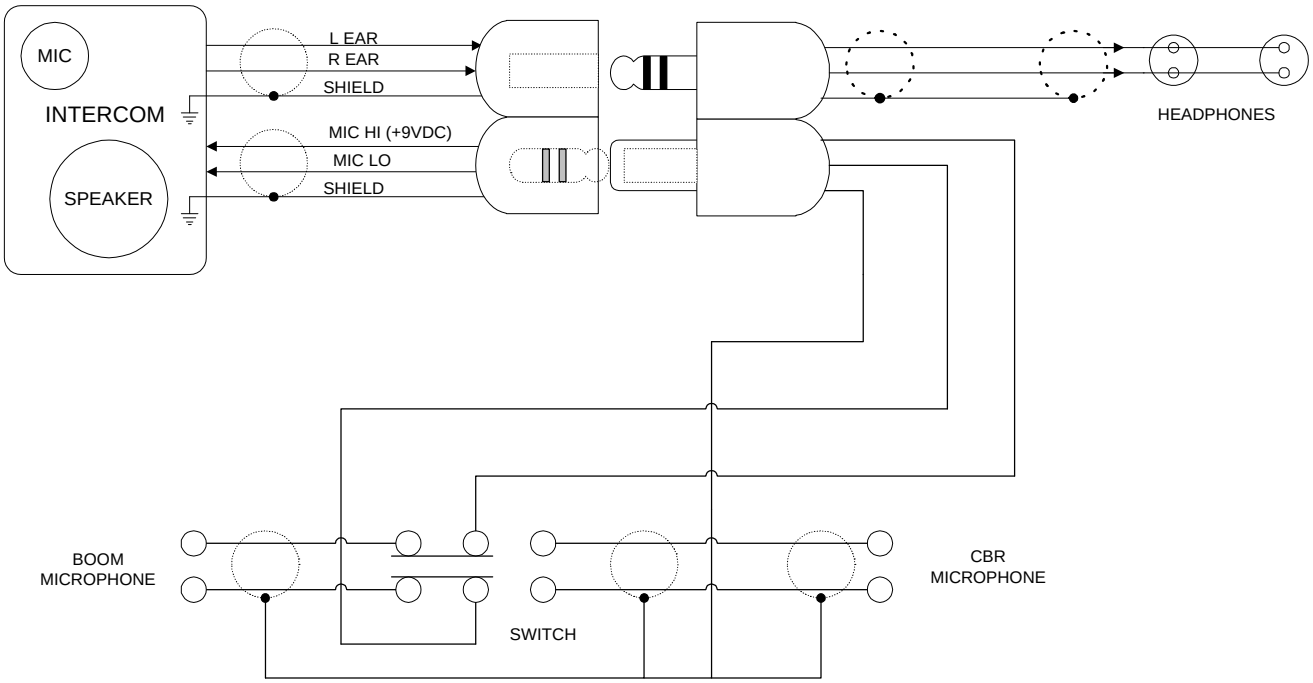


FIGURE 3. **Intercom/v-22 protective ensemble electrical interface.**

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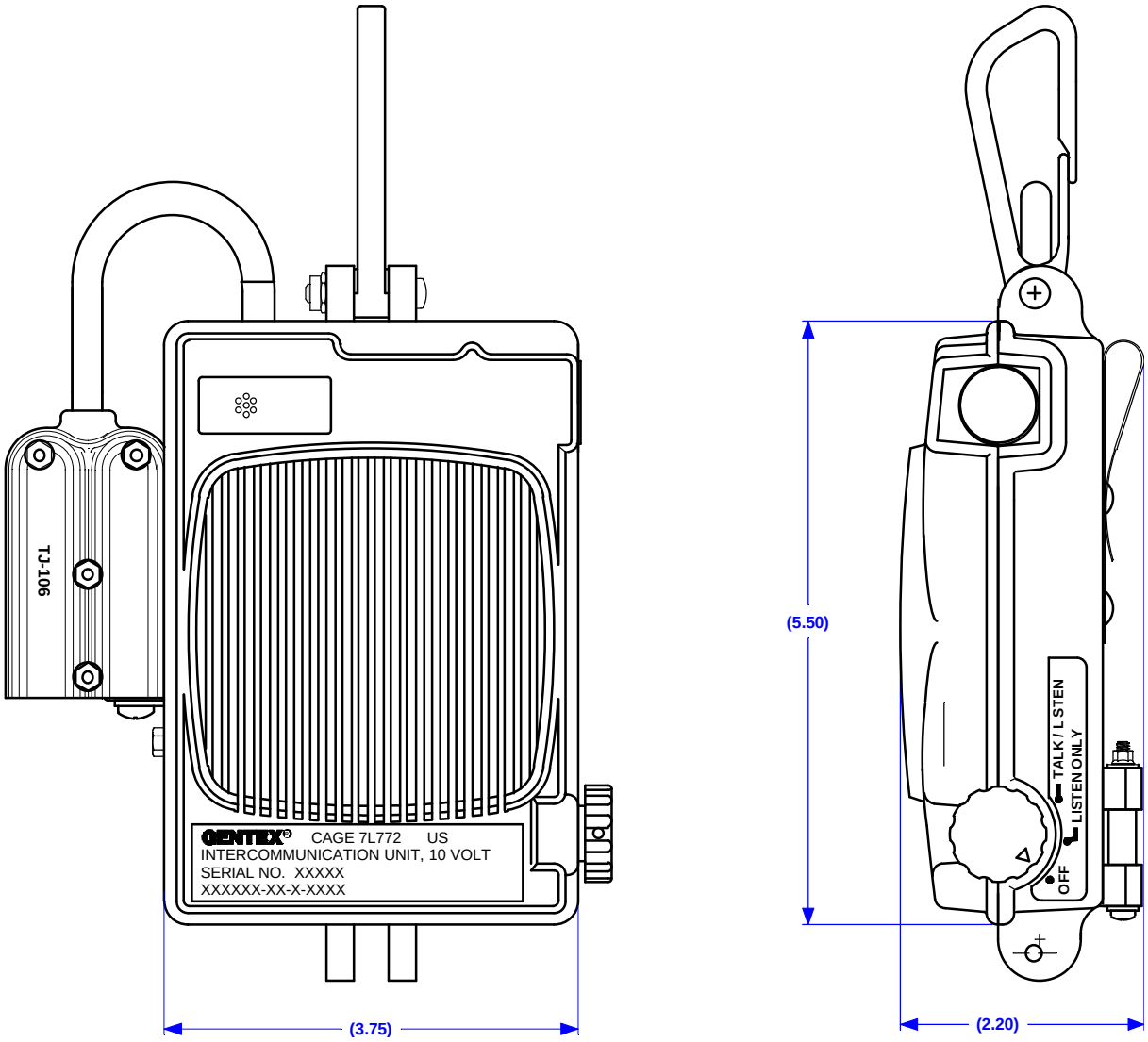


FIGURE 4. **Intercom dimensional outline.**

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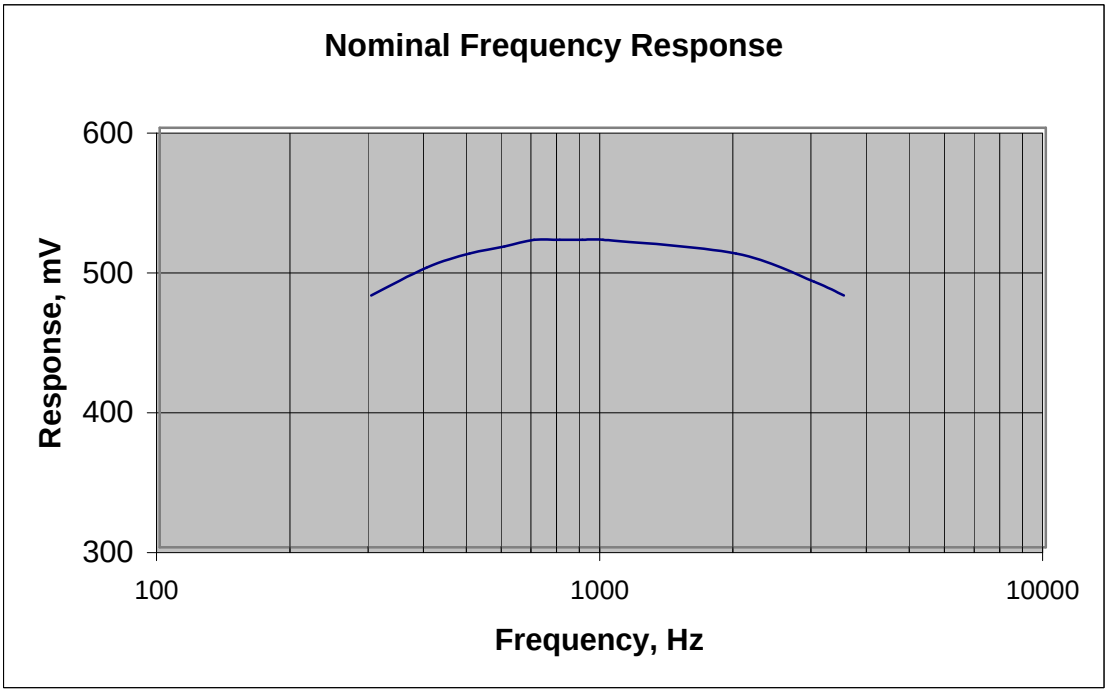


FIGURE 5. **Intercom frequency response.**

Preparing Activity:
Navy - AS
(Project Number 5895-0001)

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